James Beavan

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Summary

Junior at Georgia Tech (Mathematics, Data Science concentration) focused on data engineering. Skilled in architecting data pipelines, designing database schemas, and building backend APIs using Python, SQL, Dagster, and cloud infrastructure.

Technical Skills

Languages: Python, SQL, Java, Go
Data Engineering & ML: Dagster, DuckDB, PostgreSQL, Databricks, XGBoost, LangChain, Pandas, Tableau
Cloud, Infra & APIs: AWS (EC2, S3, Route 53), Oracle Cloud (OCI), RESTful APIs, Git, Docker, FastAPI,

Work Experience

Data Insights Intern Jun. 2026 - Aug. 2026
Northrop Grumman *Falls Church, VA*

- Developed Python scripts to automate documentation generation for 50+ SQL objects (stored procedures, views) across ~20 manufacturing sites
- Diagnosed and resolved conflicting aggregation logic across two dashboard views, aligning metric definitions and eliminating reporting discrepancies
- Optimized a core Tableau dashboard, cutting filter time by 25%
- Won CIDO Intern Hackathon with an app to track and filter international procurement data for $3M+ orders
- Built a two-stage documentation chatbot using LangChain and OpenAI with dynamic file routing and context filtering, optimizing prompts and token allocation to reduce irrelevant context

Software & Data Intern May 2025 - May 2026
Lindaben Foundation *Columbia, MD*

- Built and deployed a Dagster + DuckDB + Docker pipeline on AWS EC2 processing 180K records, with incremental loads, timestamp-based deduplication, and seven aggregation tables powering an analytics dashboard
- Built and documented 30+ REST API endpoints with filtering, pagination, and sorting, plus role-based access control across five user roles
- Developed a GPT-4-assisted migration tool converting legacy Excel delivery files into relational JSON
- Analyzed Excel time logs with Python, Pandas, and Openpyxl, detecting 300+ data entry errors and visualizing them as heat maps

Data Science Intern Sep. 2024 - Jan. 2025
Intechgambit *Falls Church, VA*

- Stress-tested a Python/Biopython-based tool for retrieving gene sequences from NCBI, identifying its upper limit at 40,000 entries before failure
- Evaluated AI models (ChatGPT, Gemini, Copilot) by testing 50+ prompts; recorded accuracy and runtime metrics to compare performance across tasks

Projects

Stockidence | stockidence.com | GitHub Aug. 2026 - Present

- Built a stock-ranking platform on Dagster and DuckDB, incrementally ingesting prices, fundamentals, and news sentiment from 4 APIs via scheduled jobs gated by per-endpoint watermarks
- Designed a three-layer DuckDB warehouse (raw → staging → mart) that derives 11 technical indicators in-pipeline instead of calling premium API endpoints
- Trained an XGBoost LambdaMART (rank:ndcg) model on 15.9K quarterly rows (377 tickers); walk-forward validated over 26 out-of-sample quarters (rank IC +0.142; top 20 averaged +4.6 pp/qtr over the S&P 500, beating it in 62% of quarters)
- Served via FastAPI + React with Docker Compose on Oracle Cloud; automated quarterly retrains

Education

Georgia Institute of Technology | Atlanta, GA Jul. 2025 - May 2028
B.S. in Mathematics, Data Science Concentration *GPA: 3.6/4.0*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Mathematics of Data Science, Statistics and Probability, Discrete Mathematics